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COURSE: CSE1402 Compiler
Design

CO4 - AT-1 class Test

Given statement

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$x = x + y * z$

else

$x = x + 1$

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Conditions: $a < b$
 $c < d$
 $a > d$

Labels: L1, L2, L3, L4, L5

L1: if $a < b$ goto L2
goto L5

L2: if $c < d$ goto L4
goto L3

L3: if $(a > d)$ goto L4
goto L5

Three Address Code

L4 $\Rightarrow x = x + y * z$

$\therefore$ L4: $t1 = y * z$
 $t2 = x + t1$
 $x = t2$
goto L6

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$$L5 \Rightarrow Z = Z + 1$$

$$\therefore L5: t3 = Z + 1$$

$$Z = t3$$

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three Address statements using Back patching

$[(a < b) \text{ AND } (c < d)] \text{ or } (e < f)$

if $(a < b)$ goto 3
goto 5

if $(c < d)$ goto 7
goto 5

if $(e < f)$ goto 7
goto 8

TRUE

FALSE

3) Consider the arithmetic expression

$$A = (-A + B) * (-A + B)$$

Three Address code:

$$t_1 = -A$$

$$t_2 = t_1 + B$$

$$t_3 = -A$$

$$t_4 = t_3 + B$$

$$t_5 = t_4 * t_2$$

$$t_6 = t_2 * t_5$$

$$A = t_6$$

OP	arg 1	arg 2	result
Unary	A	-	t ₁
+	t ₁	B	t ₂
Unary	A	-	t ₃
+	t ₃	B	t ₄
*	t ₄	C	t ₅
*	t ₂	t ₅	t ₆
=	A6	.	A

Construct the three address code using the following production for Boolean expression $(a > b)$ or $(C > d)$ or $(e < f)$:

$B \rightarrow B_1 || B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

L1% If $(a > b)$ goto LTRUE
goto L2

L2% If $(C > d)$ goto LTRUE
goto L3

L3% If $(e < f)$ goto LTRUE
goto LFALSE